

Ten-Frames: See Numbers to Ten

Answer Key

Kindergarten–Grade 1

Quick Answers

- | | |
|-------------------------------------|--------------------------------------|
| 1. bottom row = 1, dots in all = 6 | 6. dots in all = 10, empty boxes = 0 |
| 2. bottom row = 2, dots in all = 7 | 7. bottom row = 4, dots in all = 9 |
| 3. bottom row = 3, dots in all = 8 | 8. top row = 5, bottom row = 1 |
| 4. empty boxes = 1, dots in all = 9 | 9. empty boxes = 2, dots in all = 8 |
| 5. top row = 5, bottom row = 3 | 10. dots in all = 9, — |
-

What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Kindergarten–Grade 1

K.CC.B.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–4 of 5 across 20 tagged checks.

- 1.** Full top row, one dot below: bottom row and total.

Solution: The bottom row holds

1

The full top row is 5, so count on one: $5 + 1 = 6$. In all

6

- 2.** Full top row, two dots below: bottom row and total.

Solution: The bottom row holds

2

Five and two more: $5 + 2 = 7$. In all

7

- 3.** Full top row, three dots below: bottom row and total.

Solution: The bottom row holds

3

Five and three more: $5 + 3 = 8$. In all

8

- 4.** Almost-full frame: empty boxes and total.

Solution: Just one box has no dot:

1

One empty box means one less than ten: $10 - 1 = 9$. Dots in all

9

- 5.** Build 8 on an empty frame: dots per row.

Solution: Fill the top row first — it takes

5

Count on to 8: “6, 7, 8” puts three dots below, $8 - 5 = 3$. Bottom row
Check the drawing: the top row full and 3 below.

3

6. Completely full frame: total and empty boxes.

Solution: A full frame is the picture of ten — no counting needed:
Every box is used, so the empty boxes number

10

0

7. Full top row, four dots below: bottom row and total.

Solution: The bottom row holds

4

Five and four more: $5 + 4 = 9$. In all

9

8. Build 6 on an empty frame: dots per row.

Solution: The top row fills first with

5

Count on to 6: one more dot goes below, $6 - 5 = 1$. Bottom row

1

Check the drawing: full top row and a single dot below.

9. Frame with two empty boxes: empty boxes and total.

Solution: Two boxes have no dot:

2

Two less than ten: $10 - 2 = 8$. Dots in all

8

10. Challenge: full top row and 4 below — total, then tell how the frame helps.

Solution: Five and four more: $5 + 4 = 9$. In all

9

Telling how (instructor-judged): accept either frame shortcut said in the child’s own words — “the top row is 5, so I counted on 6, 7, 8, 9,” or “one box is empty, so it is one less than 10.” Counting all nine dots one by one gets the answer but is not the frame shortcut this asks for.